

Mulan C.L. Madden

Curriculum Vitae

Education

- Oct. 2025 – **Leiden University, Leiden, The Netherlands**
2029 (exp.) Ph.D. Astronomy — Supervisor: Dr. Melissa McClure, Associate Professor.
- Sept. 2023 – **Leiden University, Leiden, The Netherlands**
Jun. 2025 M.Sc. Astronomy, Specialization in Astronomy Research.
- Sept. 2018 – **Columbia University, New York, NY, USA** — **I.I. Rabi Scholar**
May 2022 B.A. Astronomy Major, Chemistry Concentration.

Research Experience

- Oct. 2025 – **Ph.D. Thesis, Leiden University, Leiden, NL** Predicting volatile and refractory depletion in the innermost region of young protoplanetary disks as an indicator of dust trapping and planetesimal formation. Models constructed with Cloudy (v25.00), a chemistry and photoionization code, will be applied to Class I, flat-spectrum, and Class II disks.
present
Supervisor: Dr. Melissa McClure, Associate Professor, Leiden University (LU).
Promotor: Dr. Michiel Hogerheijde, Associate Professor, LU.
- Sept. 2024 – **M.Sc. Thesis, Leiden University, Leiden, NL** Measured atomic NIR emission lines to constrain depletion in innermost region of the disk around T Tauri star DR Tau
Jun. 2025
Grade: 9.0/10 to probe for evidence of planetesimal formation. Investigated parameters of Cloudy (v23.01) to accurately model atomic abundances in similar young protoplanetary disks.
Supervisor: Dr. Melissa McClure, Assistant Professor, LU.
- Oct. 2023 – **M.Sc. First-Year Project, Leiden University, Leiden, NL** Studied hot HCN and C₂H₂ gas in MIR absorption toward the massive hot core binary W3 IRS5 to
July 2024
Grade: 9.5/10 derive gas temperature and density. Additionally, learned data reduction process using Spextools on raw iShell data.
Supervisors: Dr. Alexander Tielens, Professor of Physics & Chemistry, LU; Dr. Adwin Boogert, Associate Astronomer, Institute for Astronomy (IfA), University of Hawai'i (UH) at Manoa; Dr. Jialu Li, Post-Doctoral Associate, University of Maryland.
- Jan. – Aug. **National Radio Astronomy Observatory, Charlottesville, VA, USA** Investigated a protostellar outflow model and infalling streamer model constrained by
2023
observational data of molecular tracers from ALMA Large Program FAUST, for source IRAS 16544–1604 in the Bok globule CB 68.
Supervisors: Dr. Brian Svoboda, Assistant Scientist, National Radio Astronomy Observatory (NRAO), NM; Dr. Adele Plunkett, Assistant Scientist, NRAO, VA.
- July – Sept. **Center for Interstellar Catalysis, Aarhus University, Aarhus, DK** Studied
2022
CO₂ ice clusters deposited on highly oriented pyrolytic graphite (HOPG) using low temperature scanning tunneling microscopy (LT-STM). Imaged CO₂ ice clusters and analyzed data. Paper 1 of 2 submitted (see Publications).
Supervisor: Dr. Liv Hornekær, Professor of Physics & Astronomy, Aarhus University.

- Jun. 2021 – **Institute for Astronomy, University of Hawai'i at Manoa, Honolulu, HI,**
 Apr. 2022 **USA** Completed research on ice formation and grain growth in the Perseus and Serpens molecular clouds. First-author paper published in *The Astrophysical Journal*.
Supervisor: Dr. Adwin Boogert, Associate Astronomer, IfA, UH Manoa.
- May – July 2020 **Institute for Astronomy, University of Hawai'i at Manoa, Honolulu, HI,**
USA Studied ice formation and grain growth in Perseus and Serpens, observed with IRTF, Keck, and Spitzer. Spectral fits investigated 3.0 μm water ice and 9.7 μm silicate features and abundances for the dust environments and surrounding factors affecting each target source.
Supervisor: Dr. Adwin Boogert, Associate Astronomer, IfA, UH Manoa.
- June – Aug. 2019 **National Radio Astronomy Observatory, Socorro, NM, USA** Studied methanol (CH_3OH) gas transitions in the hot corino NGC 1333 IRAS 4B1 in the Perseus molecular cloud, observed with VLA, contributing to research on how grain-surface chemistry and rich dust environments give rise to planetary systems.
Supervisor: Dr. Brian Svoboda, Jansky Fellow of NRAO, NM.

Publications

- Madden, M.C.L.,** Boogert, A.C.A., Chiar, J.E., Knez, C., Pendleton, Y.J., Tielens, A.G.G.M., and A. Yip, *Infrared Spectroscopic Survey of the Quiescent Medium of Nearby Clouds: II. Ice Formation and Grain Growth in Perseus and Serpens*, 2022, *The Astrophysical Journal*, **930**, 2.
- Submitted to *Nature Communications* Kyrkjebø, S., Rønne, N., Santos, J.C., Slumstrup, L., Tang, Z., **Madden, M.C.L.,** Cassidy, A., Balog, R., Thrower, J., Ioppolo, S., Linnartz, H., Hammer, B., and L. Hornekær, *Quasicrystalline structure in low temperature two-dimensional CO_2 ice*.

Presentations

- June 2025 **Masters Colloquium, Leiden University, NL** Presented M.Sc. thesis on atomic depletion study in inner T Tauri disk DR Tau through Cloudy models.
- October 2021 **Undergraduate Research Symposium, Columbia University, NY** Recorded and presented a poster summarizing two years of research on ice formation and grain growth in Perseus and Serpens.
- October 2020 **Undergraduate Research Symposium, Columbia University, NY** Recorded a poster presentation of research on ice formation and grain growth in Perseus and Serpens and an introduction to astrochemistry and the interstellar medium, in an asynchronous event for faculty, students, and visiting parents.
- January 2020 **American Astronomical Society 235th Meeting, Honolulu, HI** Presented poster to astronomers and other conference attendees of research on methanol gas transitions in the hot corino NGC 1333 IRAS 4B1, observed with the Very Large Array at NRAO, Socorro, NM.
- October 2019 **Undergraduate Research Symposium, Columbia University, NY** Presented a poster of research on methanol gas transitions in the hot corino NGC 1333 IRAS 4B1 and gave an introduction to astrochemistry and the interstellar medium to faculty, students, and visiting parents.

- September 2019 **AstroFest, Columbia University, NY** Presented a poster at an event for Astronomy faculty and students of research on methanol gas transitions in the hot corino NGC 1333 IRAS 4B1. Explained the importance of understanding the process of grain-surface chemistry and dust-rich environments caused by hot corinos, which give rise to planetary systems.
- August 2019 **NRAO REU Program, Socorro, NM** Presented research on methanol gas transitions in the hot corino NGC 1333 IRAS 4B1, observed with the Very Large Array, in a talk to faculty and fellow summer students at the NRAO.

Teaching/Mentorship

- 2026 **Teaching Assistant, LU, Leiden, NL** TA for the Astronomy M.Sc. course High Contrast Imaging taught by Dr. Matthew Kenworthy.
- 2025 – 2026 **Supervising M.Sc. Student, LU, Leiden, NL** Daily supervisor to Ella de Vries in her first-year project modeling elemental depletion in a protostellar jet using Cloudy.
- 2021 – 2022 **Course Grader, Columbia University, NY** Graded undergraduate-level astronomy courses: Stars, Galaxies & Cosmology; Stars & Atoms; High Energy Astrophysics.

Outreach

- July 2019 **Tour Guide, NRAO, Socorro, NM** Led weekend visitor tours of the VLA.
- 2019 **Columbia Splash Educational Outreach, Columbia University, NY** Gave an introductory lecture on space exploration to New York City high school students.
- 2018 – 2019 **Columbia Space Initiative, Columbia University, NY** Student space technology and outreach club, including K-12 public events, hosting of space-related activities, NASA competitions, forums and workshops with astronauts and scientists.

Other Activities

- Spring 2022 **Course Grader** Astronomy 3273: High Energy Astrophysics, Dr. Lorenzo Sironi, Columbia University
- Fall 2021 **Course Grader** Astronomy 1836: Stars & Atoms, Dr. Kathryn Johnston, Columbia University.
- Summer 2021 **Course Grader** Astronomy 1404: Stars, Galaxies & Cosmology, Dr. Melissa Ness, Columbia University.
- 2018 – 2022 **BlueShift Astronomy Club, Columbia University, NY** As Vice President, led meetings, developed interest topics, organized presentations and events, and directed Supervisorship program.
- 2018 **NASA's Micro-g NExT, Columbia University, NY** With a group of students, worked on and presented to faculty Columbia's invention for sharp-edge detection on the International Space Station as part of NASA's annual Micro-g NExT competition.

Technical Skills

- Programming Python (advanced), IDL (basic), Java (basic), Processing (basic)
- Software LaTeX, CASA, CARTA, Gwyddion, Jupyter, Microsoft Office
- Hardware LT-STM, 3D printing, laser cutting
- OS Linux, macOS, Windows